SIMATS ENGINEERING

ASSESSMENT TOOL 1: Short Answer Test

COURSE NAME: Cloud Computing and Big Data Analytics **COURSE CODE:** CSA15

COURSE OUTCOME COVERED

CO1: *Apply cloud computing technologies including hardware-software evolution, virtualization, web services, and service models to develop cloud-based solutions.* (BL2)

Dave's Taxonomy: Imitation (Level 1) **Total Marks: 25**
Question Count: 5

Code of Conduct

I L. ASIFALI KHAN (192521009) (Name / Reg. No.)

certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.

I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the student: L. Asif

Assessment Tasks
Short Answer Test

1. Define cloud computing and explain any four characteristics that make it different from traditional on-premise computing.
2. Trace the evolution of cloud computing from hardware evolution to internet software evolution, and show how these changes enabled modern cloud platforms.
3. Explain the role of server virtualization in cloud computing. Differentiate between virtualization as an enabling technology and cloud computing as a service model.
4. Discuss the role of **data centers** in delivering cloud services..
5. Differentiate IaaS, PaaS, SaaS, and XaaS with one suitable real-world example for each model.

1) Define cloud computing and Explain any four characteristics that make it different from:.

Definition:

Cloud computing is the delivery of computing Service Such as Servers Storage database, networking and Software over the internet on demand.

Four characteristics of cloud computing:.

1) on demand Self Service:. Users can access computing resource whenever required without contacting the Service provider.

Ex: creating a virtual Machine in Azure.

2. Broad network access:. cloud Services are accessible and Smart Phones

3. Resource of Pooling:. Computing resource are Shared among Multiple users a Multiple tenant Model.

4. Rapid Elasticity:. Resource can be increased or decreased quickly according to workload requirments.

2. Trace The Evolution of cloud computing. for hardware Evolution to Internet Software Evolution and show how There changes Enabled Coudel cloud Transform:-

Cloud computing Evolved Through Several important technological development.

1. Mainframe Computer:- Early computing were large centralized computer Shared by multiple user Through Terminals.

2. Personal Computer :- Computer became Small and more affordable allowing individuals and organization to perform local computing

3. Client Server Computing:- Processing The data were Shared between client computer and centralized Server

4. Virtualization :- Multiple virtual Machine cloud a single physical Server improving Resource Utilization and Reducing cost.

5. internet development :- High Speed development Internet Enabled user to access application and computing Resource.

6. Web Services and cloud platforms:- web technologies automation and distributed computing Enabled Services to be delivered over The Internet.

3. Explain The Role of Server Virtualization in cloud Computing:-

Server virtualization: It is a technology that divides one physical Server into Multiple Independent virtual Machine using Software called Hypervisor.

Role of Server Virtualization:-

improves hardware utilization by running Multiple Machine on one Server.

Reduce hardware utilization Maintenance cost

Enable flexible allocation of CPU, Memory and Storage.

Difference between virtualization and cloud computing:-

VIRTUALIZATION	CLOUD COMPUTING.
A technology that creates virtual Resource	A Model for delivering computing Service over.
May operate with a private data center	Provides on demand access to Service and Resource
Ex: VMware esxi	Ex: Amazon EC2.

4) Discuss The Role of data centers in Cloud Services:

A data Center: is a Facility Containing Servers, Storage System Networking and Supporting Infrastructure used to Store process and manage data.

Role of data Centers in cloud Computing:

Data Storage: Store large amount of User and application and data

Processing: Provide CPU and memory resource to run application and Virtual Machine.

Networking: Servers to the internet and Transfer data between User and cloud Service

Security: protect data Through access Controls Monitoring and physical Security

Scalability: Allow cloud provide to Expand Computing Capacity According to customer demand

Example: Aws operate data Centers around The world The provide Computing and database Service to custom

5, Difference between, pass and saas with one suitable real world Example for each Model.

Cloud computing provide different Service Models have on the cloud & control and Management of the aras

Service No	FULL FORM	Description	Example
Iaas	infrastructure as a Service	Provide virtual Machine Storage and networking and Manage the as applied.	Amazon Ec2
Paas	Platform as a Service	Provide a platform and development tools to build and display application without managing the underlying infrastructure	Google app Engine.
Saas	Software as a Service	Provide Ready to use Software Through the Internet	G. Mail.
Xaas	Everything as a Service.	Deliver Ready to use various IT Resource and Capabilities as Service over network.	Sales force.

Key difference:

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Iaas:- Them have greater control over infrastructure of OS

Paas:- Developers focus Mainly on application development.

Saas:- Them access Complete Software without managing the infrastructure

Xaas:- A broad Term Covering many Service as Needed Storage.

Short Answer Test Rubric

Criteria	Weightage	Level 4 - Exemplary	Level 3 - Proficient	Level 2 - Developing	Level 1 - Beginning
Conceptual Accuracy	30%	Accurate definitions and precise explanation of all key terms	Mostly accurate with minor gaps	Partial understanding with noticeable errors	Incorrect or irrelevant explanation
Coverage of Concepts	20%	Covers all expected sub-points completely	Covers most expected points	Covers only basic points	Very limited coverage
Use of Examples	15%	Uses highly relevant cloud examples to strengthen the answer	Uses relevant examples with minor mismatch	Uses vague or partially relevant examples	No useful examples
Comparison / Differentiation	20%	Clearly distinguishes concepts with strong logic	Reasonable differentiation with minor overlap	Comparison is weak or incomplete	Fails to differentiate concepts
Clarity of Expression	15%	Clear, organized, and concise answer writing	Generally clear with few issues	Understandable but poorly organized	Difficult to follow

Total Marks: